You said:

```
import java.util.Scanner;

public class FourBitCalculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);


System.out.println("================================
=");
        System.out.println("  4-BIT BINARY CALCULATOR");

System.out.println("================================
=");

        System.out.print("Enter first 4-bit binary number: ");
        String inputA = sc.next();

        System.out.print("Enter second 4-bit binary number: ");
        String inputB = sc.next();

        if (!inputA.matches("[01]{4}") || !inputB.matches("[01]{4}")) {
            System.out.println("Invalid input! Enter exactly 4 binary
bits.");
            sc.close();
            return;
        }

        int a = Integer.parseInt(inputA, 2);
        int b = Integer.parseInt(inputB, 2);

        System.out.println("\nSelect Operation:");
        System.out.println("1. ADD");
        System.out.println("2. SUB");
        System.out.println("3. AND");
        System.out.println("4. OR");
        System.out.println("5. XOR");
        System.out.println("6. SLT (Set Less Than)");

        System.out.print("Enter your choice: ");
        int choice = sc.nextInt();

        int result;
```

```
switch (choice) {
    case 1:
        result = a + b;
        System.out.println("Operation: ADD");
        System.out.printf("Result: %4s%n",
                Integer.toBinaryString(result));
        break;

    case 2:
        result = (a - b) & 15;
        System.out.println("Operation: SUB");
        System.out.printf("Result: %4s%n",
                String.format("%4s",
                        Integer.toBinaryString(result))
                        .replace(' ', '0'));
        break;

    case 3:
        result = a & b;
        System.out.println("Operation: AND");
        printResult(result);
        break;

    case 4:
        result = a | b;
        System.out.println("Operation: OR");
        printResult(result);
        break;

    case 5:
        result = a ^ b;
        System.out.println("Operation: XOR");
        printResult(result);
        break;

    case 6:
        result = (a < b) ? 1 : 0;
        System.out.println("Operation: SLT");
        printResult(result);
        break;

    default:
        System.out.println("Invalid operation choice!");
}
```

```
        sc.close();
    }

    static void printResult(int result) {
        String binary = Integer.toBinaryString(result);
        System.out.println("Result: " +
                "0".repeat(Math.max(0, 4 - binary.length())) + binary);
    }
}
```

ChatGPT said: